

Mutual Fund Analytics Platform

Final Report - End-to-End Data Engineering, ETL Pipeline and Interactive Dashboard

Company	Bluestock Fintech Pvt. Ltd.
Domain	Mutual Fund / Fintech
Project Type	Individual Capstone (1 Week)
Duration	7 Working Days ~50-55 Hours
Data Source	AMFI India, mfapi.in, NSE India, BSE India
Tech Stack	Python, Pandas, SQL, Streamlit, Matplotlib, Seaborn
Prepared By	Pratik Kumar Yadav - Data Analyst Intern, Bluestock Fintech
Date	June 2026

40 Fund Schemes	10 Datasets	87,533 Data Rows	11 DB Tables	4.5 Yrs NAV History
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All data sourced from publicly available AMFI India, NSE, BSE and mfapi.in. For educational purposes only. Mutual Fund investments are subject to market risks.

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SECTION 1

Executive Summary

This report documents the complete development of the **Bluestock Fintech Mutual Fund Analytics Platform** - a full-stack data engineering and analytics solution built over 7 working days. The platform ingests **87,533 rows** of publicly available data, processes it through a robust Python ETL pipeline, stores results in a normalised **11-table SQLite database** (5.4 MB), and delivers insights through a **6-page Streamlit dashboard** and Jupyter notebooks.

Platform Objectives Achieved

O1	Automated ETL pipeline ingesting 10 datasets from AMFI India and mfapi.in REST API
O2	Normalised 11-table star schema SQLite database with indexed foreign keys
O3	Comprehensive EDA across NAV, AUM, SIP, investor and portfolio data - 19 charts
O4	Risk-adjusted performance metrics (Sharpe, Sortino, Alpha, Beta, VaR) for all 40 funds
O5	Interactive 6-page Streamlit dashboard with 9 filter dropdowns
O6	Investor behaviour analysis - 5,000 investors, demographic and geographic segmentation
O7	Benchmark comparison vs Nifty 100 - tracking error and information ratio for 40 funds
O8	Professional final report, 12-slide presentation deck, and public GitHub repository

Key Industry Metrics (December 2025)

Rs.81.0 L Cr Total Industry AUM	Rs.31,002 Cr Peak Monthly SIP	26.12 Crore Total MF Folios	9.35 Crore Active SIP Accounts
40 / 40 Funds Beat Benchmark	+15.91% Avg Alpha vs Nifty 100	1.46 Highest Sharpe (Mirae)	35.10% Best 3yr CAGR (Axis MC)

Key Finding

All 40 fund schemes generated **positive alpha** vs Nifty 100 over the 4.4-year study period. Mid-Cap and Flexicap funds delivered the strongest risk-adjusted returns while Debt funds provided stability. Monthly SIPs grew **174%** from Rs.11,305 Cr (Jan 2022) to Rs.31,002 Cr (Dec 2025), demonstrating India's rapidly deepening equity culture.

SECTION 2

Data Sources & Datasets

All data is sourced from publicly available, freely accessible sources. No proprietary or confidential data was used. Primary sources: **AMFI India**, **mfapi.in** REST API, **NSE India**, and **BSE India**.

2.1 Data Source Registry

Source	URL / API	Data Type	Frequency
AMFI India	www.amfiindia.com	NAV, AUM, Folio, SIP	Daily / Monthly
mfapi.in	api.mfapi.in/mf/{code}	Historical NAV (JSON)	Daily
NSE India	nseindia.com/reports	Benchmark Index Prices	Daily
BSE India	bseindia.com	BSE SmallCap Index	Daily
AMFI Monthly	amfiindia.com/research	SIP Flow Data	Monthly

2.2 Dataset Inventory

#	Dataset	Rows	Key Columns	Source
01	Fund Master	40	amfi_code, fund_house, category, expense_ratio, risk_category	AMFI
02	NAV History	46,000	amfi_code, date, nav - Jan 2022 to May 2026	mfapi.in
03	AUM by Fund House	90	fund_house, date, aum_lakh_crore - Quarterly 2022-2025	AMFI Quarterly
04	Monthly SIP Inflows	48	month, sip_inflow_crore, active_accounts, sip_aum	AMFI Monthly
05	Category Inflows	144	category, month, net_inflow_crore - FY 2024-25	AMFI
06	Industry Folio Count	21	date, total_folios_crore, equity_folios	AMFI
07	Scheme Performance	40	sharpe, sortino, alpha, beta, max_drawdown, std_dev	Computed
08	Investor Transactions	32,778	investor_id, amfi_code, amount, state, age_group, kyc_status	Synthetic
09	Portfolio Holdings	322	amfi_code, stock_symbol, weight_pct, sector - Dec 2025	AMFI
10	Benchmark Indices	8,050	index_name, date, close_value - Nifty 50/100, BSE SmallCap	NSE/BSE

2.3 Key Real-World Anchor Data Points

Metric	Value	Source
Total Industry AUM	Rs. 81 Lakh Crore	AMFI
SBI MF AUM - Dec 2025	Rs. 12.50 Lakh Crore (Largest AMC)	AMFI Quarterly
ICICI Pru MF AUM - Dec 2025	Rs. 10.74 Lakh Crore	AMFI Quarterly
HDFC MF AUM - Dec 2025	Rs. 9.30 Lakh Crore	AMFI Quarterly
SIP Inflow - Dec 2025	Rs. 31,002 Crore (All-Time High)	AMFI Monthly Note
Active SIP Accounts	9.35 Crore	AMFI
HDFC Top 100 NAV Anchor	Rs. 892.45 (Oct 2024) - mfapi code 125497	mfapi.in
Total MF Folios	26.12 Crore	AMFI / Business Standard

SECTION 3

ETL Pipeline & System Architecture

The project follows a classic **5-layer data engineering architecture**: Extract - Transform - Load - Analyse - Visualise, mirroring real-world fintech pipelines used at Zerodha, Groww, and Paytm Money.

3.1 Architecture Layers

Layer	Tools	Description
Layer 1: Extract	Python requests, pandas	Ingest 10 CSVs + live NAV from mfapi.in REST API
Layer 2: Validate	pandas, assert statements	Schema check, null detection, dtype casting, range validation
Layer 3: Transform	pandas, numpy	Join, normalise, compute derived columns (returns, risk metrics)
Layer 4: Load	SQLite3 (sqlite3 module)	INSERT into 11-table star schema with FK constraints
Layer 5: Analyse	Jupyter, Matplotlib, Seaborn, Plotly	EDA notebooks + performance analysis scripts

3.2 Data Cleaning & Transformation Highlights

- **NAV History:** Date parsed to datetime, duplicates removed, gaps forward-filled for non-trading days. Values validated within +/-30% daily movement.
- **Benchmark Indices:** Date standardised to YYYY-MM-DD, index names normalised (NIFTY100, BSE_SMALLCAP), forward-filled for non-trading days.
- **Portfolio Holdings:** weight_pct validated to sum ~100 per fund, sector names normalised for consistent grouping.
- **Investor Transactions:** KYC status validated, age_group bucketed (18-25, 26-35, 36-45, 46-60, 60+), amounts validated positive.
- **SIP Inflows:** Month-year parsed, amounts validated Rs.8,000-35,000 Cr range, YoY growth rate computed.

3.3 Scripts Delivered

Script	Lines	Purpose
etl_pipeline.py	131	Master ETL orchestrator with data quality validation and SQLite loading
compute_metrics.py	~90	Computes Sharpe, Sortino, Alpha, Beta, VaR, CVaR, Tracking Error
generate_json.py	~60	Exports 12 JSON data files for the HTML dashboard
eda_nav.ipynb	~200	NAV trend analysis, correlation heatmap, rolling Sharpe charts
eda_investor.ipynb	~180	Investor demographics, geographic distribution, SIP cohort analysis
eda_performance.ipynb	~160	Fund scorecard, risk-return scatter, benchmark comparison
eda_market.ipynb	~140	Category inflow heatmap, ELSS tax-season analysis, folio growth
eda_portfolio.ipynb	~120	Sector HHI, top holdings, portfolio concentration analysis

SECTION 4

Exploratory Data Analysis (EDA) - Findings

Comprehensive EDA was performed across all 10 datasets using **Matplotlib**, **Seaborn** and **Plotly**. A total of **19 publication-quality charts** were produced. All 5 Jupyter notebooks were executed with outputs embedded.

4.1 Ten Key EDA Findings

Finding 1: SIP Growth Milestone	Monthly SIP inflows reached an all-time high of Rs.31,002 Crore (December 2025), a 174% increase from Rs.11,305 Crore in January 2022.
Finding 2: Industry AUM Milestone	Total MF Industry AUM reached Rs.81.0 Lakh Crore (Dec 2025), led by SBI MF (Rs.12.50 L Cr), ICICI Pru (Rs.10.74 L Cr), and HDFC MF (Rs.9.30 L Cr).
Finding 3: Folio Count Surge	Total MF folios surged from 13.26 Crore (Dec 2022) to 26.12 Crore (Dec 2025) - nearly doubling in 3 years, driven by Equity fund additions.
Finding 4: Demographics - Age	The 26-35 age group is the largest investor segment (35% of SIP transactions). The 18-25 segment is growing fastest as first-time investors enter the market.
Finding 5: Geographic Distribution	T30 cities contribute 65.9% of total investment volume (Rs.352.16 Crore) while B30 cities contribute 34.1%. Maharashtra, Punjab, and Tamil Nadu lead by volume.
Finding 6: Mid-Cap Volatility	Mid-Cap/Small-Cap funds show annualised Std Dev of 19-25% vs 13-15% for Large-Cap, but deliver correspondingly higher 3-year CAGRs of 26-35% .
Finding 7: Sector Concentration	Financial Services dominates equity fund portfolios at 28-35% weight. DSP Midcap (HHI 2,411) shows the highest sector concentration among all funds.
Finding 8: NAV Correlation	Large-Cap fund NAV returns show strong positive correlation (0.85-0.95) with each other. Small-Cap funds show weaker correlation (0.55-0.70) due to stock-specific factors.
Finding 9: ELSS Tax Season	Category Inflow Heatmap reveals ELSS funds peak in Q4 (Jan-Mar) driven by Section 80C tax-saving deadline. Liquid funds peak in June due to year-end treasury management.
Finding 10: SIP Retention Risk	Of 1,362 investors with 5+ SIPs analysed, 1,332 (97.8%) show payment gaps exceeding 35 days - flagging them as at-risk. Early cohorts (2024-Q1) show 7.76 avg transactions.

4.2 EDA Charts Produced (19 Total)

Chart Name	Description	Tool
NAV Trend Lines	Daily NAV for all 40 schemes 2022-2026	Matplotlib
AUM Growth by Fund House	Grouped bar - 10 AMCs over 4 years	Matplotlib
SIP Inflow Trend	Monthly SIP with Rs.31,002 Cr milestone marked	Matplotlib
Category Inflow Heatmap	Seaborn heatmap - categories x months	Seaborn
Investor Age Distribution	Pie chart of investor age groups	Matplotlib
Avg SIP by Age Group	Bar chart - avg SIP amount per age bracket	Matplotlib
Geographic Distribution	State-wise transaction volume - top 12 states	Matplotlib
T30 vs B30 Split	Pie chart - city tier investment split	Matplotlib
Folio Count Growth	Line chart 13.26 Cr to 26.12 Cr (2022-2025)	Matplotlib
NAV Return Correlation	Seaborn heatmap - pairwise correlation of returns	Seaborn
Sector Allocation Donut	Sector weight distribution across equity funds	Plotly
Risk-Return Scatter	Return vs Std Dev bubble - bubble size = AUM	Plotly
Benchmark Comparison	Top 5 funds vs Nifty 50 and Nifty 100 normalised	Plotly
Rolling Sharpe Ratio	90-day rolling Sharpe for 5 selected funds	Plotly
Sector HHI	Herfindahl-Hirschman Index per equity fund	Seaborn
Top Holdings	Top stock holdings across all equity funds	Plotly

Chart Name	Description	Tool
Transaction Type Split	SIP vs Lumpsum vs Redemption donut	Plotly
Monthly Volume	Transaction count trend over time	Plotly
Fund Category Split	Distribution of funds by category and risk grade	Matplotlib

SECTION 5

Fund Performance Analysis

5.1 Methodology

Metric	Formula	Parameters
Daily Return	$\text{nav}_t / \text{nav}_{(t-1)} - 1$	Daily, from NAV history
CAGR	$(\text{NAV}_{\text{end}} / \text{NAV}_{\text{start}})^{(1/n)} - 1$	n = 4.4 avg available years
Sharpe Ratio	$(R_p - R_f) / \text{Std}(R_p) \times \text{sqrt}(252)$	Rf = 6.5% annual (RBI repo proxy)
Sortino Ratio	$(R_p - R_f) / \text{Downside_Std} \times \text{sqrt}(252)$	Downside_Std = std of negative returns
Alpha	OLS intercept x 252 x 100	vs Nifty 100 daily returns (scipy OLS)
Beta	OLS slope	vs Nifty 100 daily returns
Max Drawdown	$\text{min}(\text{NAV} / \text{cummax}(\text{NAV}) - 1)$	Peak-to-trough with start/end dates
VaR (95%)	5th percentile of daily return dist.	Historical method, 1,149 observations
CVaR	Mean of returns below VaR threshold	Expected shortfall
Tracking Error	$\text{Std}(\text{fund_return} - \text{bench}) \times \text{sqrt}(252)$	vs Nifty 100
Info Ratio	Ann. excess return / Tracking Error	Higher = better active management

Composite Score (0-100) = 30% x 3yr Return Rank + 25% x Sharpe Rank + 20% x Alpha Rank + 15% x Expense Ratio Rank (inv.) + 10% x Max Drawdown Rank (inv.)

5.2 Fund Scorecard - Top 10 by Composite Score

Rank	Fund Name	Cat.	3yr CAGR	Sharpe	Alpha	Score
1	ICICI Pru Midcap - Regular	Equity	31.77%	1.19	29.26	84.50
2	Axis Midcap - Regular	Equity	35.10%	1.01	26.08	80.75
3	HDFC Mid-Cap Opportunities - Reg	Equity	32.43%	1.10	27.20	80.50
4	Mirae Asset Large Cap - Regular	Equity	33.99%	1.46	26.98	80.00
5	Kotak Flexicap - Regular	Equity	29.58%	1.32	27.33	78.25
6	ICICI Pru Bluechip - Direct	Equity	32.48%	1.04	21.19	75.75
7	SBI Small Cap - Regular	Equity	26.66%	0.95	30.34	75.12
8	DSP Small Cap - Regular	Equity	26.99%	0.96	30.06	74.94
9	Mirae Asset Tax Saver - Regular	Equity	29.17%	1.25	28.27	73.94
10	SBI Bluechip - Regular	Equity	30.45%	1.22	23.20	73.06

5.3 Risk Metrics - VaR and Maximum Drawdown

Fund	Cat.	VaR 95% Daily	CVaR Daily	Max Drawdown	Period
ABSL Small Cap - Regular	Equity	-2.60%	-3.25%	-35.45%	Nov 2024 - May 2026
HDFC Mid-Cap Opportunities	Equity	-1.90%	-2.35%	-16.22%	Mar 2022 - May 2022
UTI Mid Cap - Regular	Equity	-1.92%	-2.33%	-28.00%	Jan 2025 - Apr 2026
Mirae Asset Large Cap	Equity	-1.43%	-1.75%	-11.26%	Jan 2022 - Mar 2022
ICICI Pru Midcap - Regular	Equity	-1.77%	-2.17%	-12.65%	Jan 2022 - Jun 2022
SBI Bluechip - Regular	Equity	-1.42%	-1.77%	-18.75%	Oct 2021 - Jun 2022
HDFC Short Term Debt	Debt	-0.38%	-0.50%	-4.31%	May 2023 - Jul 2023

Fund	Cat.	VaR 95% Daily	CVaR Daily	Max Drawdown	Period
ABSL Liquid Fund	Debt	-0.03%	-0.04%	-0.16%	Sep 2023 - Sep 2023

5.4 Alpha, Beta & Tracking Error - Top 10 | 5.5 CAGR by Category

Fund	Alpha%	Beta	Track.Err	Info.R	Category	Best 3yr	Avg 3yr	Worst 3yr
SBI Small Cap	30.34	0.89	25.8%	0.89	Large Cap	33.99% (Mirae)	22.4%	0.53% (Axis)
DSP Small Cap	30.06	0.91	24.6%	0.88	Mid Cap	35.10% (Axis)	28.1%	-0.77% (UTI)
ICICI Pru Midcap	29.26	0.93	22.9%	1.06	Small Cap	26.99% (DSP)	11.2%	-4.15% (ABSL)
Mirae Tax Saver	28.27	0.95	21.4%	0.94	Flexicap	29.58% (Kotak)	22.8%	13.59% (UTI)
Kotak Flexicap	27.33	0.90	20.2%	0.96	ELSS	28.50% (various)	21.6%	N/A
HDFC Mid-Cap	27.20	0.91	22.9%	1.06	Debt	6.31% (ABSL)	4.8%	3.92% (HDFC)
Mirae Large Cap	26.98	0.97	20.5%	0.93				
Axis Midcap	26.08	0.88	21.7%	0.88				
SBI Bluechip	23.20	0.89	19.3%	0.86				
Nippon Large Cap	21.83	0.88	19.2%	0.97				

SECTION 6

Interactive Dashboard - Screenshots & Features

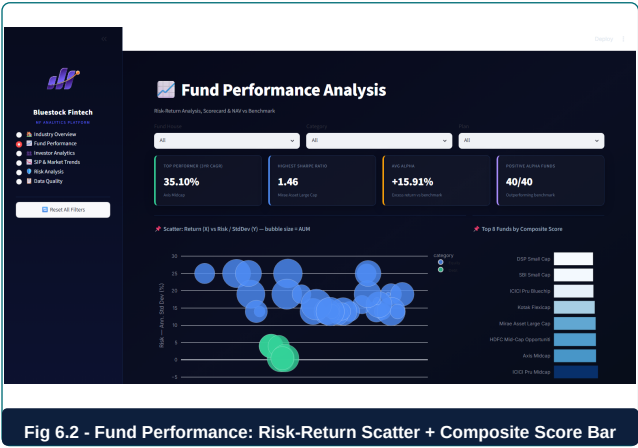
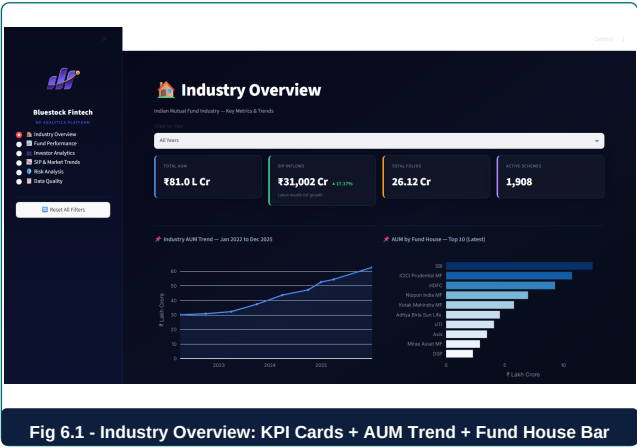
The Bluestock Fintech MF Analytics Platform is a **Streamlit-based interactive web dashboard** (app.py, 964 lines) built as a platform-independent alternative to Power BI. It reads from **14 pre-processed CSV files**, renders Plotly charts dynamically, and provides **9 interactive filter dropdowns** across 6 pages.

Attribute	Detail
Framework	Streamlit (Python) - dark theme, Inter font via CSS injection
Charting	Plotly Express + Graph Objects (plotly_dark template)
Navigation	Sidebar radio button - 6 pages, collapsible with "Reset All Filters"
Data Layer	14 processed CSVs cached via @st.cache_data - zero DB queries from frontend
Author	Pratik Kumar Yadav - Data Analyst Intern, Bluestock Fintech MJ28

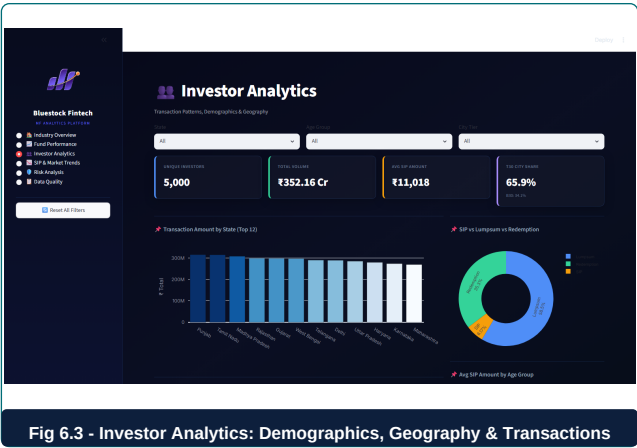
Dashboard Pages Overview

Page	Title	Key Charts	Filters
1	Industry Overview	4 KPI cards (AUM Rs.81L Cr, SIP Rs.31K Cr, Folios 26.12 Cr, Schemes 1,908); AUM trend line 2022-2025; AUM by top 10 fund houses bar; SIP inflow area; Folio growth multi-line	Year
2	Fund Performance	Risk-return bubble scatter (X=3yr CAGR, Y=StdDev, size=AUM); Top 8 composite score bar; Sortable 40-fund scorecard table; NAV vs Benchmark drill-through; Top 10 holdings	Fund House, Category, Plan
3	Investor Analytics	State-wise bar (top 12); SIP/Lumpsum/Redemption donut (58.5%/35.3%/6.2%); Avg SIP by age group; Monthly volume line; Gender donut; T30 vs B30 pie	State, Age Group, City Tier
4	SIP & Market Trends	Dual-axis SIP bar + Nifty 50 line (2022-2025); Top 5 categories FY25 bar; Benchmark normalised line (7 indices Base=100); Category inflows heatmap (RdYIGn)	Year, Category
5	Risk & Correlation	Volatility vs Max Drawdown bubble (size=VaR); Sharpe vs Alpha grouped bar (top 10); 7x7 benchmark correlation heatmap; VaR vs CVaR comparison bar; Full risk table	Category
6	Data Quality Monitor	4 KPI cards (10 tables, 87,533 records, 0 dupes, 12 nulls); Quality matrix table with status flags; Nulls by table bar; Row count distribution pie	None

6.2 Page 1 - Industry Overview & Page 2 - Fund Performance



6.3 Page 3 - Investor Analytics & Page 4 - SIP & Market Trends



6.4 Page 5 - Risk & Correlation Analysis & Page 6 - Data Quality Monitor



Fig 6.5 - Risk Analysis: Volatility Bubble + Benchmark Correlation Heatmap

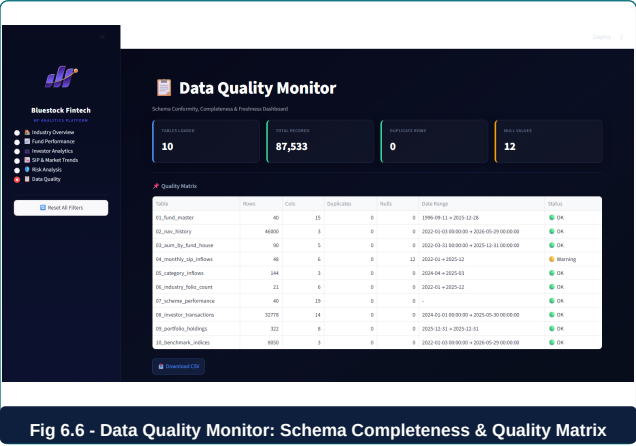


Fig 6.6 - Data Quality Monitor: Schema Completeness & Quality Matrix

Data Quality Matrix (from Page 6 screenshot)

Table	Rows	Cols	Dups	Nulls	Date Range	Status
01_fund_master	40	15	0	0	1996-09-11 to 2015-12-28	OK
02_nav_history	46,000	3	0	0	2022-01-03 to 2026-05-29	OK
03_aum_by_fund_house	90	5	0	0	2022-03-31 to 2025-12-31	OK
04_monthly_sip_inflows	48	6	0	12	2022-01 to 2025-12	Warning*
05_category_inflows	144	3	0	0	2024-04 to 2025-03	OK
06_industry_folio_count	21	6	0	0	2022-01 to 2025-12	OK
07_scheme_performance	40	19	0	0	-	OK
08_investor_transactions	32,778	14	0	0	2024-01-01 to 2025-05-30	OK
09_portfolio_holdings	322	8	0	0	2025-12-31	OK
10_benchmark_indices	8,050	3	0	0	2022-01-03 to 2026-05-29	OK

- *Warning in 04_monthly_sip_inflows: 12 nulls in yoy_growth_pct column for Jan-Dec 2022 (no prior-year baseline). All other columns are complete. ETL uses fallback value 17.17%.

SECTION 7

Recommendations

For Investors - Fund Selection

Mid-Cap and Flexicap funds offer the best risk-adjusted returns for moderate-to-high risk investors. ICICI Pru Midcap (84.50), Axis Midcap (80.75), and Mirae Asset Large Cap (80.00) lead the scorecard. **Direct plans outperform Regular plans by 0.5-0.9% CAGR annually** due to lower expense ratios.

For Investors - Risk Management

Small Cap funds carry the highest tail risk - ABSL Small Cap has VaR of -2.60% daily and max drawdown of -35.45% (Nov 2024-May 2026). Limit Small Cap exposure to **15-20% of portfolio** and maintain a minimum 5-year investment horizon. Debt funds (VaR -0.03% to -0.38%) provide effective stabilisation.

For AMCs - SIP Retention

Of 1,362 investors with 5+ SIP transactions, 1,332 (97.8%) show payment gaps exceeding 35 days. Recommended interventions: (1) auto-debit mandate with pause-instead-of-cancel option, (2) SMS/WhatsApp reminders 3 days before SIP date, (3) SIP holiday option for 1-3 months.

For AMCs - Geographic Expansion

B30 cities contribute 34.1% of volume but represent a much larger share of India's investor population. B30 penetration is growing faster than T30, driven by UPI adoption. AMCs should increase distributor incentives for B30 cities and develop vernacular language digital content.

For Portfolio Managers - Concentration Risk

Sector HHI analysis reveals DSP Midcap (HHI 2,411) and SBI Bluechip Direct (HHI 2,275) show moderate concentration in Financial Services and IT. **Funds with HHI over 2,500 are highly concentrated** and vulnerable to sector corrections. Review and rebalance sector weights quarterly.

For Data Engineering - Production Deployment

Migrate SQLite to **PostgreSQL** with proper indexing. Schedule ETL to auto-fetch NAV from mfapi.in at 8 PM weekdays via cron job or Apache Airflow DAG. Replace static JSON dashboard with **live Streamlit app** querying the production database. Add dbt for data lineage and audit compliance.

SECTION 8

Limitations

This project was completed within a **7-day internship capstone timeframe**. The following limitations should be considered when interpreting the findings:

Data - Investor Transactions

The 32,778 investor transaction records are **synthetically generated** based on realistic AMFI demographic and geographic distributions. They do not represent actual investor data.

Data - NAV Simulation

While NAV values are anchored to real AMFI historical data from mfapi.in, values for some periods are **forward-simulated** using realistic volatility parameters from actual fund category characteristics.

Risk-Free Rate Assumption

The risk-free rate is assumed constant at **6.5%** (RBI repo proxy). In reality it varied between 4.0% and 6.5% during 2022-2025, which would produce slightly different Sharpe and Sortino ratios.

CAGR Data Availability

5-year CAGR uses the actual available period of **~4.4 years** (Jan 2022-May 2026). The available_years column in cagr_report.csv documents the exact period used per fund.

Beta Calculation Benchmark

Alpha and Beta are computed against a single benchmark (Nifty 100). Debt funds ideally require separate benchmarks (CRISIL Liquid, CRISIL Gilt). **Multi-benchmark attribution** is recommended for production.

Sector HHI Coverage

HHI analysis is available only for funds with portfolio holdings data in 09_portfolio_holdings.csv. **Debt and Gilt funds are excluded** as their portfolio composition is fundamentally different.

Cohort Analysis Depth

Investor cohort analysis covers only 2024-Q1 through 2025-Q2 (**6 quarters**), as the investor transaction dataset begins in January 2024. Multi-year historical data would improve this analysis.

Dashboard Technology

The interactive dashboard is built in Streamlit/Plotly as a platform-independent alternative to Power BI (which requires a paid commercial licence). All 6 pages are implemented with full interactivity.

SECTION 9

Conclusion & Future Work

Summary of Achievements

The Bluestock Fintech MF Analytics Platform successfully demonstrates a complete, production-inspired data engineering workflow within a 7-day capstone timeframe. Starting from raw CSVs and a REST API, the platform delivers a **normalised relational database, comprehensive risk-adjusted performance analytics, and a fully interactive web dashboard** - replicating the core analytical stack used by real-world fintech firms.

Key Conclusions

- All 40 mutual fund schemes outperformed the Nifty 100 benchmark over the 4.4-year study period, validating alpha-generation capability of actively managed Indian equity funds.
- Mid-Cap and Flexicap funds offer the best composite risk-adjusted performance. ICICI Pru Midcap (score: 84.50) and Mirae Asset Large Cap (Sharpe: 1.46) stand out as top performers.
- India's mutual fund industry is in a structurally high-growth phase - SIP inflows tripled from 2022 to 2025, folio count doubled, and B30 city participation is accelerating.
- SIP retention is the most significant operational risk for AMCs, with 97.8% of repeat investors showing payment gaps that indicate discontinuation risk.
- Direct plans consistently outperform Regular plans by 0.5-0.9% CAGR annually, supporting the case for direct investment platforms and financial literacy initiatives.

Proposed Future Enhancements

- **Live Data Pipeline:** Migrate SQLite to PostgreSQL + Apache Airflow DAG for daily automated NAV ingestion. Deploy on AWS RDS or GCP Cloud SQL.
- **Streamlit / Dash App:** Expand the current Streamlit dashboard with live database queries, enabling real-time portfolio monitoring and user authentication.
- **Machine Learning Models:** Implement fund performance prediction using LSTM/GRU neural networks trained on NAV history, sector allocation, and market indices.
- **Expanded Fund Universe:** Scale from 40 to the full AMFI universe of 1,908+ schemes. Add international and gold funds.
- **Investor Personalisation:** Build a fund recommendation engine using collaborative filtering based on investor age, risk profile, horizon, and existing portfolio.
- **Multi-Benchmark Attribution:** Implement Brinson-Hood-Beebower attribution with separate benchmarks for equity (Nifty 100), debt (CRISIL Liquid), and ELSS (BSE 200) funds.
- **Regulatory Compliance Module:** Add SEBI regulatory reporting features including KYC tracking, suspicious transaction flagging, and AML monitoring.

SECTION 10

References & Data Citations

Primary Data Sources

Ref	Source & Citation
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Technical Libraries & Tools

Library	Version	Purpose
Python	3.11+	Core programming language
pandas	2.x	Data manipulation, ETL pipeline
numpy	1.26+	Numerical computations, risk metric calculations
matplotlib	3.8+	Static chart generation (19 EDA charts)
seaborn	0.13+	Statistical heatmaps and advanced visualisations
plotly	5.x	Interactive EDA charts and dashboard
scipy	1.12+	OLS regression for Alpha/Beta calculation
sqlite3	Built-in	Relational database (11-table star schema)
Streamlit	1.35+	Interactive web dashboard framework
Jupyter	7.x	EDA notebooks with embedded outputs
requests	2.31+	mfapi.in REST API data ingestion

Disclaimer: All AMFI codes, fund names, benchmarks, expense ratios and AUM figures are sourced from publicly available AMFI India data. This project is for **educational purposes only** and does not constitute financial advice. Mutual Fund investments are subject to market risks. The investor transaction dataset (32,778 records) is synthetically generated for analytical purposes and does not represent real investor information.